# Tumble Pop

## Programming Fundamentals Project Rubric (Fall 2025) [300+100]

**Programming Fundamentals Project Self-Evaluation**

**Student 1 Name: _____**

**Roll Number: _____**

**Section: ___**

**Student 2 Name: _____**

**Roll Number: _____**

**Section: ___**

**Programming Fundamentals Concepts (30 Marks)**

| Sr # | Feature | Total Marks | Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 1 | Proper use of functions, arrays, and static memory allocation. Code must be modular with clear separation of concerns. | 30 | | |

**Core Game Mechanics (35 Marks)**

| Sr# | Feature | Total Marks | Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 2 | Character Selection (Yellow/Green TumblePopper with unique abilities) | 3 | | |

| | | | | |
|---|---|---|---|---|
| 3 | Vacuum Device Implementation (W A S D directional control - 4 directions) | 5 | | |
| 4 | Enemy Capture Mechanism (capacity: 3 in Level 1, 5 in Level 2) | 5 | | |
| 5 | Single Shot & Vacuum Burst | 7 | | |
| 6 | Projectile Movement in Four Directions with Floor Logic | 5 | | |
| 7 | Collision Detection | 10 | | |

**Enemy System (35 Marks)**

| Sr# | Feature | Total Marks | Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 8 | Ghost Enemy | 4 | | |
| 9 | Skeleton Enemy | 6 | | |
| 10 | Invisible Man Enemy | 7 | | |
| 11 | Chelnov Enemy | 10 | | |
| 12 | Enemy Defeat Logic | 3 | | |

**Level Design & Platform Mechanics (25 Marks)**

| Sr# | Feature | Total Marks | Assigned Marks | Evaluation Marks |
|---|---|---|---|---|
| 13 | Level 1: Straight platforms with 8 Ghosts & 4 Skeletons | 10 | | |
| 14 | Level 2: Random generation and slanted platforms with sliding mechanics | 8 | | |
| 15 | Level 2: Sequential enemy spawning (3 Invisible Man, | 7 | | |

| | | | | |
|---|---|---|---|---|
| | 4 Chelnov, 9 Skeletons, 4 Ghosts) | | | |

**Power-ups & Health System (15 Marks)**

| Sr# | Feature | Total Marks | Assigned Marks | Evaluation Marks |
|---|---|---|---|---|
| 16 | Speed Power-up (Doubles character speed) | 3 | | |
| 17 | Range Power-up (Increases suction angle) | 3 | | |
| 18 | Power Power-up (Increases suction radius) | 3 | | |
| 19 | Extra Life Power-up | 3 | | |
| 20 | Health System (3 lives per level, instant death on enemy touch, resets on new level) | 3 | | |

**Scoring System (15 Marks)**

| Sr# | Feature | Total Marks | Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 21 | Enemy Capture Points | 4 | | |
| 22 | Combat Bonuses (Multi-Kill, Aerial Defeat, Vacuum Burst) | 3 | | |
| 23 | Combo Multiplier | 3 | | |
| 24 | Level Clear Bonuses | 3 | | |
| 25 | Penalty System | 2 | | |

# PHASE 2 [150 Marks]

### Prerequisites (30 Marks)

| Sr # | Feature | Total Marks | Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 26 | Proper use of pointers, dynamic memory allocation, and memory management (new/delete). All Phase 1 features must be retained and functional. Proper resizing logic and no memory leaks. | 30 | | |

### Advanced Mechanics (45 Marks)

| Sr# | Feature | Total Marks | Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 27 | Yellow TumblePopper: 360° Rotation System (W A S D keys, smooth angle control) | 5 | | |
| 28 | Dynamic Enemy Capture with Memory Resizing (No capacity cap, exact allocation, proper new/delete) | 8 | | |
| 29 | Projectile Release at Angle of Elevation (based on 360° rotation) | 10 | | |
| 30 | DynamEnemies Producer (Runtime enemy generation until destroyed, dynamic allocation) | 10 | | |
| 31 | Moving Cloud Platform (Up/down movement, holds producer, becomes platform when producer destroyed) | 7 | | |

## Boss Level (50 Marks)

| Sr# | Feature | Total Marks | Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 32 | Octopus Boss with Dynamic Minion Spawning (Continuous production until destroyed) | 10 | | |
| 33 | Random Tentacle System with Dynamic Resizing (Random position, random duration) | 10 | | |
| 34 | Minion-Tentacle Interaction (Minion rebounds/respawns on tentacle hit) | 15 | | |
| 35 | Enemy Following Mode (Activated when boss at 2 HP, real-time player tracking) | 5 | | |
| 36 | Dynamic resizing of level (number of rows and cols are increased by 1.5 and cell size is decreased appropriately) | 10 | | |

## UI and QOL Features

| Sr# | Feature | Total Marks | Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 37 | Pause & Reload System | 10 | | |
| 38 | Main Menu | 5 | | |
| 39 | Leaderboard System | 5 | | |

**Advanced Power Up**

| Sr# | Feature | Total Marks | Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 40 | Ball Breaker Power-up (Wall reflection, split into 2 balls with different speeds) | 10 | | |

**Bonus Features (only awarded if all other features are implemented)**

| Sr # | Feature | Total Marks | Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 41 | Pixel Perfect Hit Boxes | 10 | | |
| 42 | Boss Angry State (At 2 HP: water rises(using perlin noise), minions grow, appearance/color changes) | 20 | | |
| 43 | Boss Level Tile Destruction Power-up (7-10 seconds, activated by specific combo) | 10 | | |
| 45 | Rewind Token | 15 | | |
| 46 | Soft Tiles in Boss Level (Sinking mechanics, 5 second delay, 20% sink threshold doubles speed) | 15 | | |
| 47 | BlackOut Mode | 20 | | |
| 48 | Level 0: Interactive Tutorial (Teaches controls through level design, enforced actions) | 10 | | |

**Deductions**

| Sr# | Feature | Total Marks | Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 49 | Use of global variables | -20 | | |
| 50 | Poor Code structure / No modularity | -20 to -40 | | |
| 51 | Use of Dynamic memory allocation | -30 | | |
| 52 | Plagiarism Detected | **F** | | |
| 53 | No Code Documentation/ No Comments | -10 | | |
| 54 | Game Crashes of Major Bugs | -10 to -25 | | |
| 55 | Memory Leak | -20 | | |
| 56 | Poor Dynamic Memory management | -20 | | |

Note:

- The Evaluation Sheet must be hand filled.
- Failure to bring the project rubrics to the demo will incur a 30-mark deduction.